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The cognitive-motor coordination dataset: gait analysis under combined blindfold, virtual reality-induced cognitive load, and vibrotactile feedback

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ABSTRACT

How cognitive load and impairments in visual perception affect gait quality and how vibrotactile feedback can substitute high cognitive load and impaired vision in such conditions are two fundamental research questions in the fields of motor control and cognitive science. Answering these questions can provide us with a better understanding of our cognitive-motor coordination and help us to design cognitive human-machine interfaces (C-HMI) that can improve the gait control and balance in elderly and visually impaired individuals. However, there is still no general and publicly available dataset to accelerate the studies in this field. Thanks to the recent technological developments in virtual reality (VR), we designed an experiment to analyze gait during treadmill walking with variable speeds under four different scenarios, including (1) blindfold without vibrotactile feedback, (2) blindfold with vibrotactile feedback, (3) VR-induced cognitive load without vibrotactile feedback, and (4) VR-induced cognitive load with vibrotactile feedback. Twenty-eight healthy individuals (12 female and 16 male) participated in this experiment. For each participant and scenario, data were recorded from multiple sources, including videos from three viewpoints (top, side, and rear), questionnaire responses, ground reaction forces (GRF), treadmill speed, foot sensor data, head positions and orientations captured by VR headset, and the experimental performance scores, including game, instability, and positioning scores. The presented dataset can be used for different research purposes, such as studying the impact of visual impairments on gait quality, the possibility of improving proprioception by vibrotactile feedback, analyzing the coordination between human cognition and neuromuscular system, examining the human strategies for instability management, identifying talented athletes, and assessing the gait control and balance strategies in visually impaired individuals.

Background & Summary

Gait control is a complex mechanism of our cognition and neuromuscular system that draws the attention of cognitive scientists and motor control researchers^{1,2}. Based on the scientific evidence, humans rely on cognitive attention³, visual perception⁴, and proprioception⁵ for gait and balance control. Therefore, visual impairment, visual-motor coordination issues, and high cognitive load result in a drastic drop in gait quality and increase the falling risk, which is the case in elderly individuals⁶.

Based on WHO reports, by 2030, one in six people worldwide will be aged 60 years or older⁷, and as of 2023, one in four people worldwide was visually impaired due to blindness, refractive error, and age-related issues⁸. Among this growing population, falling is one of the most common and serious health concerns, due to but not limited to visual impairment and cognitive load^{9–11}, with approximately one in three elderly individuals experiencing a fall each year¹². These unintentional falls are the leading cause of injury or death in older adults, often resulting in sedentary behavior, long-term disability, chronic pain, and decreased quality of life^{13–15}. Furthermore, these injuries impose economic burdens on healthcare systems and governments¹⁶. Consequently, gait assessment during instability conditions and fall prevention have become a critical public health and research priority.

To effectively prevent falls, it is essential to study the mechanism of human balance and stability, which requires a comprehensive gait dataset. Early studies have highlighted the role of coordinated hip and ankle angles in maintaining postural balance^{17,18}. Subsequent work has built on this foundation with detailed gait datasets collected at multiple walking speeds, capturing kinematic, kinetic, and electromyographic data^{19–21}. Recently, researchers have used immersive and non-immersive virtual reality (VR) to examine how it influences gait and postural stability^{22,23}.

Despite these efforts, several gaps remain in publicly available datasets. First, existing datasets rarely include experiments involving variable walking speeds, uncertain conditions, cognitive load, blindfold walking, and external perturbations. This is important for understanding how our neuromuscular and cognitive system deal with such difficulties. Second, with the growing use of augmented, virtual, and mixed reality in daily life and clinical rehabilitation, there is limited assessment of its effect on gait stability and its great potential to create realistic instability or cognitive load conditions. The third limitation is the absence of conditions simulating visual impairment (e.g., walking with eyes closed). For instance, some balance-control datasets include standing with eyes open or closed²⁴, but locomotor vision-restricted conditions are lacking in gait datasets.

An additional gap is the integration of vibrotactile feedback as a simple solution for gait quality improvement and fall prevention, which has shown potential for improving postural and gait stability. Studies have demonstrated that wearable biofeedback devices can increase stride length by providing real-time feedback on swing time²⁵. Additionally, vibrotactile cues have been effective in reducing postural sway during standing²⁶ and walking²⁷. Vibrotactile cues can also contribute to increase the gait symmetry in disable elderly individual during walking²⁸. These findings suggest that vibrotactile feedback can be effective for maintaining stability under visually or cognitively demanding conditions.

To address these gaps, we present a new multimodal gait dataset collected under four experimental scenarios: (1) closed-eye (blindfold) treadmill walking without vibrotactile feedback, (2) closed-eye (blindfold) treadmill walking with vibrotactile feedback, (3) VR-induced cognitive load treadmill walking without vibrotactile feedback, and (4) VR-induced cognitive load treadmill walking with vibrotactile feedback. This design enables examination of gait dynamics and the coordination of the neuromuscular system, cognition, and sensory feedback on stability strategies. Data were collected from 28 participants, each performing a 7-minute trial per scenario. The dataset includes synchronized video recordings from three viewpoints (top, side, and rear), data from an instrumented treadmill (ground reaction forces, foot sensor, and treadmill speed), and head kinematics (head position and orientation in three axes; X , Y , Z) captured by the VR headset.

This dataset can be used to answer fundamental research questions such as the impact of visual impairment and cognitive load on gait quality/control and the possibility of using vibrotactile cues for improving gait control and stability. It can also be used to design simple and effective cognitive human-machine interfaces (C-HMI) for fall prediction and prevention, as well as analyzing the coordination between human cognition and the neuromuscular system, examining human strategies for instability management, identifying talented athletes, and developing C-HMI for navigation of visually impaired individuals.

Methods

Participants

Twenty-eight healthy individuals (12 females and 16 males, age 22.04 ± 2.03 years, weight 69.04 ± 12.36 kg, and height 173 ± 8.18 cm; mean $\pm$ s.t.d.) without neuromuscular, physical, or cognitive impairments participated in this experimental study. The participants' overall demographic and physical information is presented in Fig. 1.

All participants provided written consent after being informed about the purpose and procedures of the experiment and their right to withdraw at any time. The study was conducted in accordance with the Declaration of Helsinki and approved by the Research Ethics Committee of the Faculty of Physical Education and Sport Sciences—University of Tehran (IR.UT.SPORT.REC.1403.103).

To incentivize task engagement and compensate participants for their time, we implemented a performance-based reward structure. All participants received monetary compensation, and the top three performers (based on overall task scores) received an additional prize.

Experiment setup

The experimental setup contains four subsystems, as shown in Fig. 2: (1) treadmill, (2) virtual reality (VR) headset, (3) camera and image processing, and (4) vibrotactile feedback.

The treadmill subsystem is an instrumented, programmable device produced by h/p/cosmos for medical purposes (Gaitway II S, h/p/cosmos, Germany). In addition to a measurement system—comprising two force plates, a speed sensor, a foot sensor, and data acquisition software (Gaitway, v2.07, KISTLER)—this subsystem includes a user interface (para graphics 2.6.14 software, h/p/cosmos) that allows the operator to generate various treadmill patterns^{29,30}. In total, 112 unique speed patterns were generated for each participant and scenario, each consisting of a randomized combination of four phases: increasing speed, decreasing speed, fixed speed, and stopped intervals³¹. Across these patterns, the treadmill speed varied from 0% to 140% of each participant's preferred walking speed, with a constant acceleration of 1.17 m/s^2 for both ramping up and down. An exemplary generated pattern is shown in Fig. 3. To ensure safety, two experimenters were always positioned behind and beside the treadmill, and all participants wore a height-adjusted safety harness that prevented falls without restricting natural walking.

The VR headset subsystem, the Meta Quest 3, was used for data collection, recording data from its internal inertial measurement unit (IMU) sensor, including head position (X , Y , Z) and head orientation (X , Y , Z) during the experiment. It

was also used for running the cognitive games to incorporate user's cognitive load, attention disruption, and visual perception disturbances into the experiment.

The developed game contained time-limited mathematical questions, including solving summations and subtractions, completing a math series, and counting specific objects in pictures; see Fig. 4. The time for responding to mathematical and object counting questions were 5 and 7 seconds, respectively. To increase the cognitive and physical challenge and prevent participants from adapting to the experimental conditions, the questions' position was made dynamic, meaning that they were not located statically at a specific place during the experiment. Each question was randomly placed within the 80-degree field of view on an imaginary sphere in front of the participants' faces. In addition, the questions' position was set relative to the participant's head. If the participant moved to the end of the treadmill, the questions moved accordingly. This ensured that the user couldn't use the relative position of the questions to adapt the treadmill position. The cognitive game's relative positions (X , Y , Z) are also recorded.

For each question, the player receives +1 for a correct answer, and -1 for either an incorrect or a missed response. Accordingly, the overall score is computed as $game\ score = N_c - N_w - N_m$, where N_c , N_w , and N_m are the numbers of correct, wrong, and missed answers, respectively. The scores acquired during the experiment were also recorded.

The camera and image processing subsystem consists of three video-recording cameras positioned to capture top, side, and rear views of the treadmill. In addition to video recording, this subsystem uses a Python script to process the frames captured by the top-view camera and to extract the 2D position (X , Z) of the participant on the treadmill by detecting a marker attached to a belt around the participant's waist (located 1 meter above the treadmill surface with 30 cm length), referred to as the marker belt. This subsystem uses a control law to generate vibration cues by comparing the center position of the marker with the center of the predefined safe zone (see Fig. 5a). As shown in this figure, the predefined safe zone is defined with dimensions extending 8 cm laterally to the left/right, 14 cm anteriorly, and 16 cm posteriorly, relative to the center of the treadmill. The spatial distribution of the marker center positions for all participants across all scenarios is visualized using contour plots (contourf function in MATLAB) in Fig. 5b. The proportion of time individuals remain within the safe zone is also calculated and defined as the center score.

The vibrotactile feedback subsystem consists of four similar Eccentric Rotating Mass (ERM) vibration motors, each positioned on a direction of the participants' abdomen (forward, right, backward, and left) via a belt, called vibration belt. The motors are connected to motor driver modules (TB6612FNG), which regulates the supplied voltage between 0 and 12 volts. The vibration intensity of each motor is controlled using pulse width modulation (PWM) method, where a pulse width of 0(255) corresponds to the minimum (maximum) terminal voltage. The motor drivers are interfaced with an ESP-WROOM-32 microcontroller, which enables wireless communication via Wi-Fi. The ESP32 module receives control commands through a Python-based interface, allowing real-time adjustment of motor activation and vibration intensity. The vibration motors are activated based on the control law generated by the camera and image processing subsystem.

The overall experiment protocol includes two sessions conducted on separate days, as shown in Fig. 6. The first day corresponds to the familiarization session, and the second day to the main experiment session. Regardless of the session, the experiment consists of four scenarios, each involving 7 minutes of treadmill walking: blindfold without vibrotactile feedback (BFNF), blindfold with vibrotactile feedback (BFWF), VR-induced cognitive load without vibrotactile feedback (CLNF), and VR-induced cognitive load with vibrotactile feedback (CLWF). The choice of 7-minute trials was based on the common use of trial duration ranging from 5 to 20 minutes in gait studies^{29,32,33}. In addition, based on the pilot experiments 7-minute trial duration satisfied sufficient exposure to treadmill perturbations while avoiding participant mental and physical fatigue during whole experiment; i.e., four 7-minute scenarios. The randomly generated treadmill patterns are considered as external disturbances/perturbations that along with VR-induced cognitive load or blindfold scenarios can lead to gait distortion or instability.

Familiarization session

The familiarization process includes 13 steps; see Fig. 6 : (1) collecting personal information presented in Fig. 1, (2) explaining the experiment procedure by showing a prerecorded video, (3) tutoring the participant on how to play the cognitive game with the VR headset (off the treadmill), (4) warming up the participant through treadmill walking, (5) determining the preferred walking speed: the participant walks on the treadmill starting from a low speed (approximately 2 km/h), and the speed is increased in steps of 0.3 km/h up to 4.5 km/h, with each step lasting 15 seconds. The participant chooses the preferred speed while walking by comparing the current speed with the next and previous ones, and the extracted preferable walking speed is used as the input for the randomly generated treadmill patterns, (6) familiarizing the participant with the treadmill walking pattern generated based on the preferred speed, (7) attaching the marker belt at a height of 1 meter above the treadmill surface, followed by attaching the vibration belt and ensuring each vibration motor is correctly positioned, (8) familiarizing the participant with the vibration feedback function, (9) familiarizing the participant with BFNF scenario (repeated as needed), followed by rest, (10) familiarizing the participant with BFWF scenario (repeated as needed), followed by rest, (11) familiarizing the participant with CLNF scenario (repeated as needed), followed by rest, (12) familiarizing the participant with CLWF scenario (repeated as

needed), followed by rest, (13) filling out the questionnaire form to make participants familiar with questions (responses are not saved).

Main experiment session

The main experiment session begins with warming up the participant by walking on the treadmill. The participant then completed the four scenarios (BFNF, BFWF, CLNF, and CLWF) in a randomized order, walking for 7 minutes in each, with a randomly generated treadmill pattern, followed by an approximately 5 – 10 minutes rest and answering the related questionnaire questions; see Fig. 6. After completing the fourth scenario and answering to its corresponding questionnaire questions, the participant reviewed their entire responses to the questionnaire and modified them if needed.

Data cleaning

The data cleaning consisted of 3 steps: (1) Resampling all data to 200 Hz, since the sampling rate of treadmill, VR headset, camera and image processing subsystems are different; 1000 Hz, 72 Hz, and 60 Hz, subsequently; (2) Transforming data units, including treadmill measured voltages (V) to force (N) and speed (m/s), participant location from pixels to (X , Z) coordinates; (3) Shifting and rotating head positions and orientations to align with the camera and image processing subsystem coordinates.

Data Records

The dataset includes data recorded during the main experiment session from four subsystems: treadmill, VR headset, camera and image processing, and vibrotactile feedback subsystems. Detailed information about each collected data, along with its source, unit, and description is presented in Table 1. The recorded videos of the side-view camera can be used to extract the joint angles in the sagittal plane using for instance MediaPipe library³⁴, as shown in Fig. 7. The availability of the extraction code, along with examples, is described in the Code Availability section.

The dataset titled **The cognitive-motor coordination dataset: gait analysis under combined blindfold, virtual reality-induced cognitive load, and vibrotactile feedback** consists of tabular-structured files organized across six folders and is available via a figshare repository³⁵. In the **Cleaned Data** directory, each participant's cleaned data from all measurements is stored in a single file for each experimental scenario. The **Raw Data** directory contains the unprocessed data captured from the camera and image processing and the VR headset subsystems. Notes reporting any issues or key points that occurred during the experiment for each participant are stored in the **Notes** directory. The **Treadmill Raw Data** directory includes treadmill-measured data, separated by participant and experiment scenario. The **Video** directory contains recordings from three viewpoints (top, side, and rear), organized by participant and scenario. The **Metadata & Questionnaire Responses** directory stores participants' metadata, including preferred speed, scenario order, gender, age, height, weight, waist circumference, dominant hand and foot, regular sport activity, regular treadmill usage, and previous experience with a VR headset. This directory also includes participants' reviewed questionnaire responses. The hierarchical structure of the dataset is illustrated in Fig. 8.

Questionnaire

The questionnaire contains various questions, with possible answers being distinct numbers from 0 to 10, asking participants to rate different parameters related to the experiment, including difficulty, comfort, safety, stability, the similarity of treadmill perturbations to real-world instabilities, fear of instability, physical and cognitive effort/load, which were answered by the participant after each scenario. For the mentioned questions, 0(10) corresponds to minimum(maximum) similarity to the normal gait or real-world condition. For instance in question "How safe did you feel during this stage of the experiment?" a score of 0(10) corresponds to the minimum(maximum) perceived safety, similar to the participant's normal gait. For the scenarios involving vibrotactile feedback, additional questions addressed the effectiveness of feedback in improving stability, reducing fear of instability, reducing physical and cognitive effort/load, as well as the participant's attention to and trust in feedback. For these questions, 0(10) corresponds to the minimum(maximum) effectiveness, attention, and trust. The questionnaire questions and the summarized rating results are shown in Fig. 9.

Technical Validation

To assess the quality, consistency, and reliability of the collected dataset, several validation procedures were performed using different subsets of the data, each addressing a specific aspect of the experimental protocol. The validation included visual inspection of the recorded measurements, statistical analysis of experiment assumptions, and normality assessment of both one-dimensional (1D) and two-dimensional (2D) variables. Unless otherwise specified, statistical analyses were conducted with a significance level of 0.05, and Bonferroni correction was applied when multiple pairwise comparisons were performed. All analyses were carried out using MATLAB (R2023b, MathWorks Inc., USA) and SPSS (Version 26, IBM Corp., USA).

Normality testing

Normality analyses were additionally performed on the released dataset to characterize its statistical properties and to guide the selection of appropriate statistical methods for subsequent analyses. Because the dataset contains both one-dimensional (1D) and two-dimensional (2D) variables, different normality tests were employed according to the dimensionality of the data. For 1D variables, including VR position (per axis), center score, and solving time, the *Shapiro–Wilk*³⁶, *D’Agostino–Pearson*³⁷, and *Kolmogorov–Smirnov*³⁸ tests were applied. For 2D variables, including the marker overall position coordinates (X, Z) presented in Fig. 5b, multivariate normality was evaluated using *Mardia’s multivariate normality test*³⁹ and the *Henze–Zirkler’s consistent test*⁴⁰.

Data grouping and validation subsets

The validation process was organized into two complementary parts.

Validation 1, VR position distribution

This subset consists of the head position recorded by the VR headset during each 7-minute walking scenarios. For each participant, the head position along the three spatial axes (X, Y, Z) was averaged over time, resulting in 12 representative values per participant (3 axes $\times$ 4 scenarios). The resulting distributions are presented as boxplots in Fig. 10, providing a visual assessment of the variability across participants and the stability of the VR tracking system throughout the experiment.

Validation 2, experimental protocol

To verify that the experimental protocol itself did not introduce systematic performance changes over time, participant performance was evaluated using three complementary measures calculated for each one-minute interval of the 7-minute task: game score, center score, and solving time (Fig. 11). The game score and the center score, correspond to the VRScore and POSScore signals in the dataset, are described in Table. 1. Solving time represents the time spent by subjects successfully solving questions from the moment the question was instantiated until they provided a response.

The normality of each measure was first assessed using the *Shapiro–Wilk* test to determine the appropriate statistical analysis. For paired comparisons, all possible comparisons between the seven 1-minute intervals were performed (21 comparisons in total) with a significance threshold of $\alpha = 0.05$; the Bonferroni correction was also considered for multiple comparisons.

The game score satisfied the normality assumption and was therefore analyzed using paired-sample *t*-tests. Statistically significant differences were observed only between the first minute and some of the subsequent intervals (5th, 6th, and 7th), whereas no significant differences were found among any pair of the remaining six minutes. These results indicated a brief change at the beginning of the experiment, after which game performance remained stable. The center score did not satisfy the normality assumption and was consequently analyzed using the paired *Wilcoxon signed-rank* test. After the analysis, none of the pairwise comparisons reached statistical significance after Bonferroni correction ($p \geq 0.060$). Solving time did not satisfy the normality assumption. Since participants completed a different number of questions during each minute, the observations could not be paired. Therefore, comparisons were performed using the unpaired *Mann–Whitney U* test, where no statistically significant differences were observed for solving time ($p \geq 0.109$).

These findings indicate that the selected game difficulty and experiment duration were appropriately balanced. The cognitive task was sufficiently challenging to prevent measurable learning effects while remaining manageable enough to avoid fatigue-related performance degradation, as evidenced by the stable performance across most time intervals. Consequently, learning and fatigue can be considered negligible confounding factors, supporting the validity of the experimental protocol and the interpretation that the observed responses primarily reflect the investigated experimental conditions.

Visualization and data consistency

Visual inspection was performed to verify the integrity and consistency of the collected data. Boxplots (Fig. 10 and Fig. 11) were used to identify potential outliers, assess the spread of measurements, and examine the distribution of each variable under different experimental conditions or time periods. Additionally, the scatter plot of participants’ positions on the treadmill across all scenarios, illustrated in Fig. 5, indicates that all participants properly attended to the experimental goal by trying to stay within the safe zone. It can clearly be seen that the density of points within the safe zone is higher across the treadmill. These visualizations complement the statistical analyses by providing an intuitive overview of the variability and consistency of the recorded measurements.

Summary

Overall, the validation results demonstrate that the dataset is internally consistent, statistically reliable, and suitable for subsequent quantitative analyses. The combination of visual inspection, normality assessment, and validation of the experimental protocol confirms both the quality of the recorded measurements and the stability of the results throughout the experiment. These analyses provide confidence that the dataset can serve as a robust benchmark for future studies on vibrotactile feedback, virtual reality, and gait analysis.

Usage Notes

The dataset comprises three types of files: MP4 files for video recordings, GAUX files for treadmill raw data, TXT for notes, and CSV files for all other data. The MP4 files can be viewed using any standard video player. The GAUX, TXT, and CSV files are encoded in UTF-8 and can be accessed and processed as tabular data in MATLAB or as data frames using the Python Pandas library with the specified encoding.

Data Availability

The complete data repository is publicly available on Figshare repository (<https://doi.org/10.6084/m9.figshare.c.8584895>).

Code Availability

The code for extracting joints angles in the sagittal plane from side-view video recordings, along with examples, are available in the Gait Joints Angles Extraction directory³⁵.

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Author Contributions

Mohammad Sajjad Alamdar: experiment design, data collection, cleaning dataset, literature review, writing the paper, preparing figures. Kamyar Rahmani: experiment design, data collection, and proofreading the text. Mansour Davoudi: data collection, cleaning dataset, writing the paper, preparing figures, and statistical analysis. Diba Ravanshid: data collection, writing the paper, preparing figures, and proofreading the text. Rezvan Nasiri: supervision, experiment design, literature review, writing the paper, and data illustration. Majid Nili Ahmadabadi: supervision, experiment design, and editing the paper.

Competing Interests

The authors declare no competing interests.

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Figures and tables

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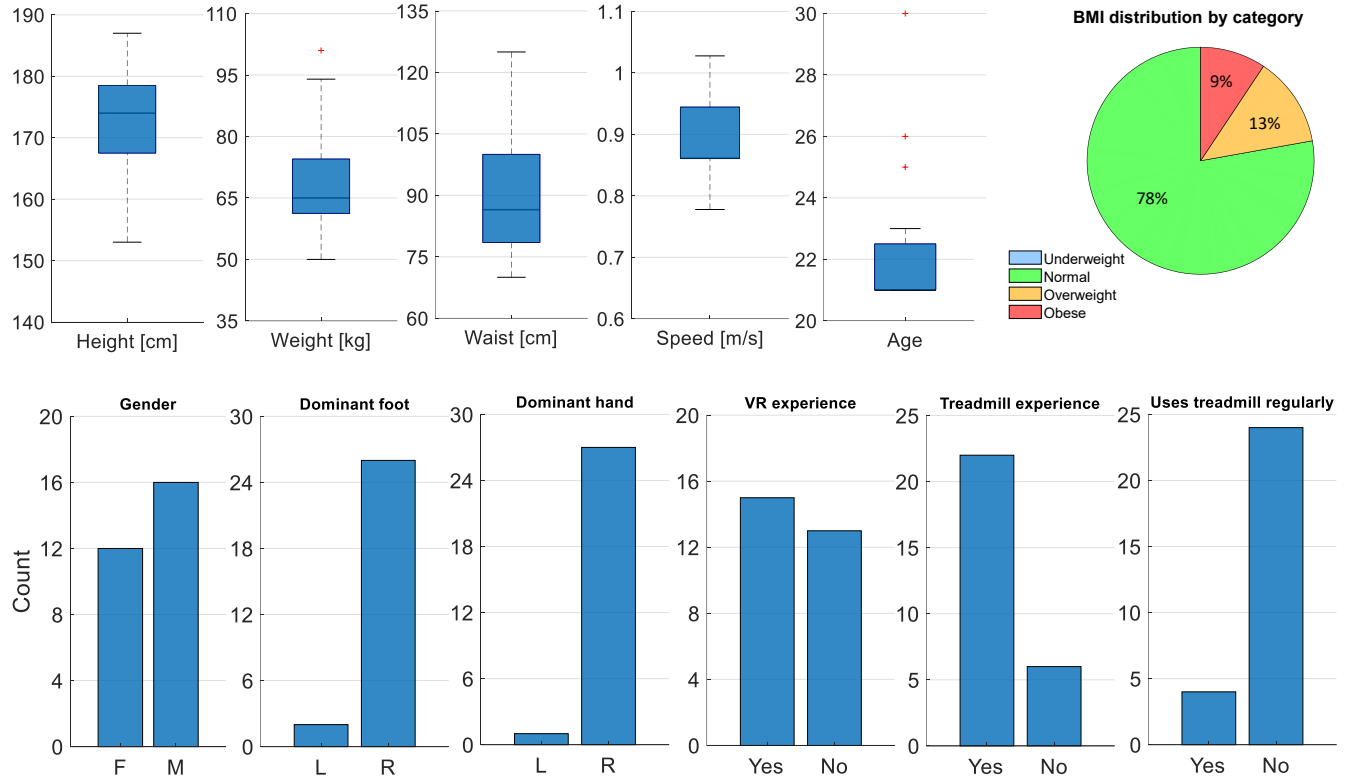


Figure 1. Overview of participants' demographic and physical information, along with their prior experience with the virtual reality headset and frequency of treadmill use. The data are presented using bar plots, box plots, and a pie chart.

Table 1. Description, source, and units of extracted variables used in the experiment.

Variable	Source	Unit	Description
Time	System clock	s	Time stamp of each recorded sample
ForcePlate1–8	Force plates	N	Ground reaction force data from force plates 1 to 8
SpeedSensor	Treadmill speed	m/s	Measured treadmill or walking speed
FootSensor	Foot sensors	V	Foot contact or gait phase detection sensor
CameraPixelZ	Top-view camera	m	Position of the tracked marker along the treadmill's longitudinal (forward–backward) direction
CameraPixelX	Top-view camera	m	Position of the tracked marker along the treadmill's lateral (side-to-side) direction
VibroLeft	Vibrotactile belt	duty cycle	Raw PWM activation level of the left-side vibrotactile motor
VibroBackward	Vibrotactile belt	duty cycle	Raw PWM activation level of the backward vibrotactile motor
VibroRight	Vibrotactile belt	duty cycle	Raw PWM activation level of the right-side vibrotactile motor
VibroForward	Vibrotactile belt	duty cycle	Raw PWM activation level of the forward vibrotactile motor
AwarenessScore	Outside flags	–	The number of times participants reached the front or rear treadmill edge (signal values = +1 and –1, respectively).
POSScore	Camera tracking	–	Score representing how long the participant's marker remained within the safe zone of the treadmill
VRPosX, VRPosY, VRPosZ	VR headset	m	Position of the participant in the virtual environment (X, Y, Z axes)
VRRotX, VRRotY, VRRotZ	VR headset	degrees	Orientation (rotation) of the participant in the virtual environment (X, Y, Z axes)
ObjectPosX, ObjectPosY, ObjectPosZ	VR environment	m	Position of the reference object in the virtual environment (X, Y, Z axes)
VRScore	VR environment	–	Score representing the participant's game performance in the VR environment during the experiment

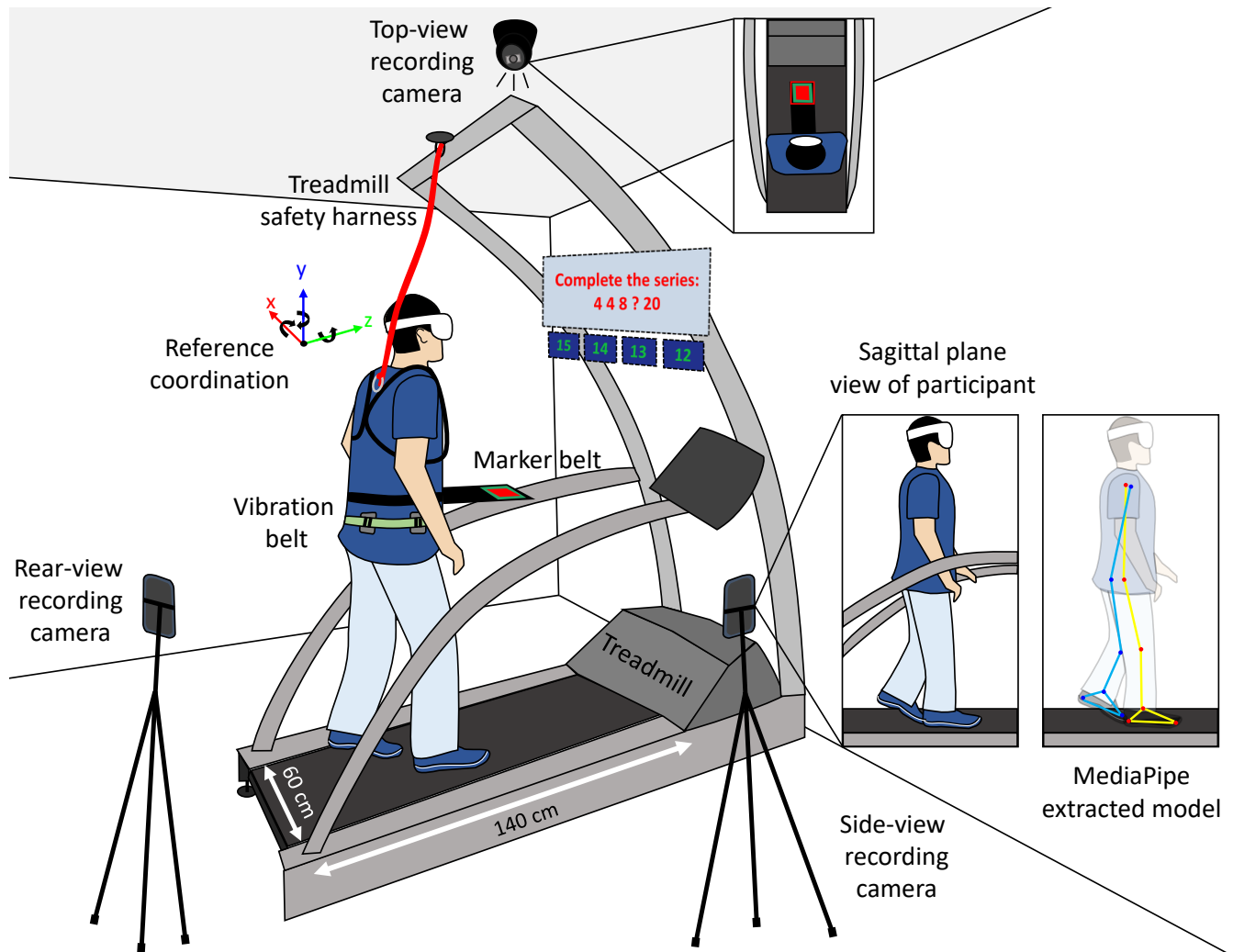


Figure 2. This figure depicts a participant in the experimental setup, which consists of four subsystems: (1) the **treadmill subsystem**, generating walking patterns and collecting force, speed, and foot information; (2) the **VR headset subsystem**, rendering the cognitive game and recording head kinematics (position and orientation), questions position, and game score; (3) the **camera and image processing subsystem**, comprising three cameras (top, side, and rear views) used for recording videos, extracting the joint angles in the sagittal plane, and generating instability cues for the vibrotactile subsystem attached to the participant; (4) the **vibrotactile feedback subsystem**, consisting of a marker and four vibration motors attached to the participant via their corresponding belts. Motors are activated according to the control law presented in Figure 5 from the camera and image processing subsystem. The marker size was considered large enough to make sure that the top-view-camera can easily detect it. The marker's center represents the participant's position on the treadmill.

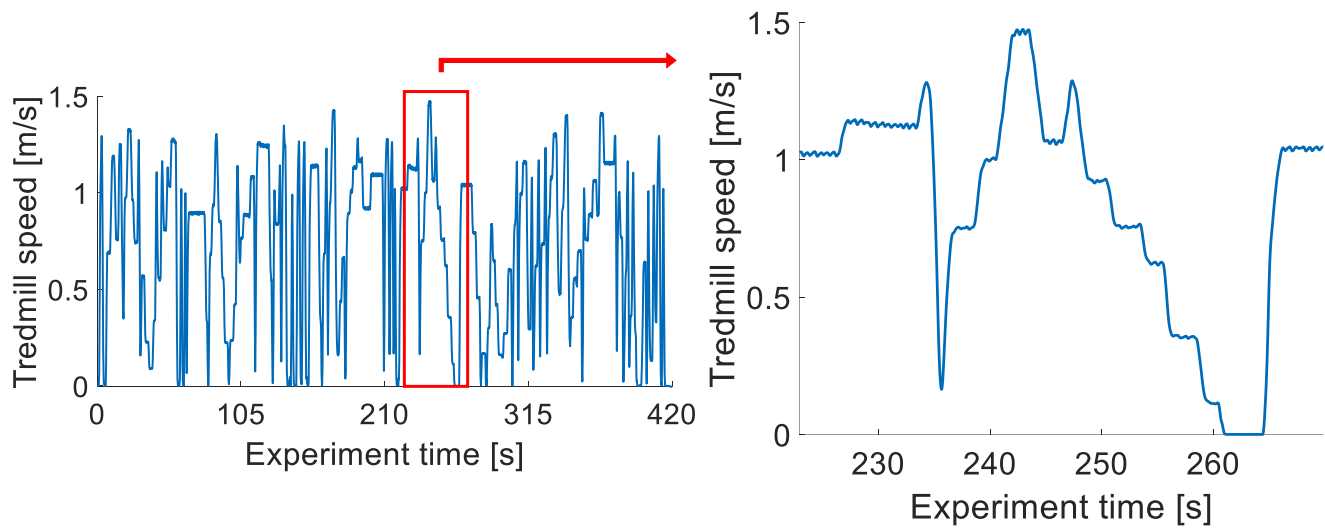


Figure 3. A randomly generated treadmill speed pattern recorded by treadmill speed sensor.

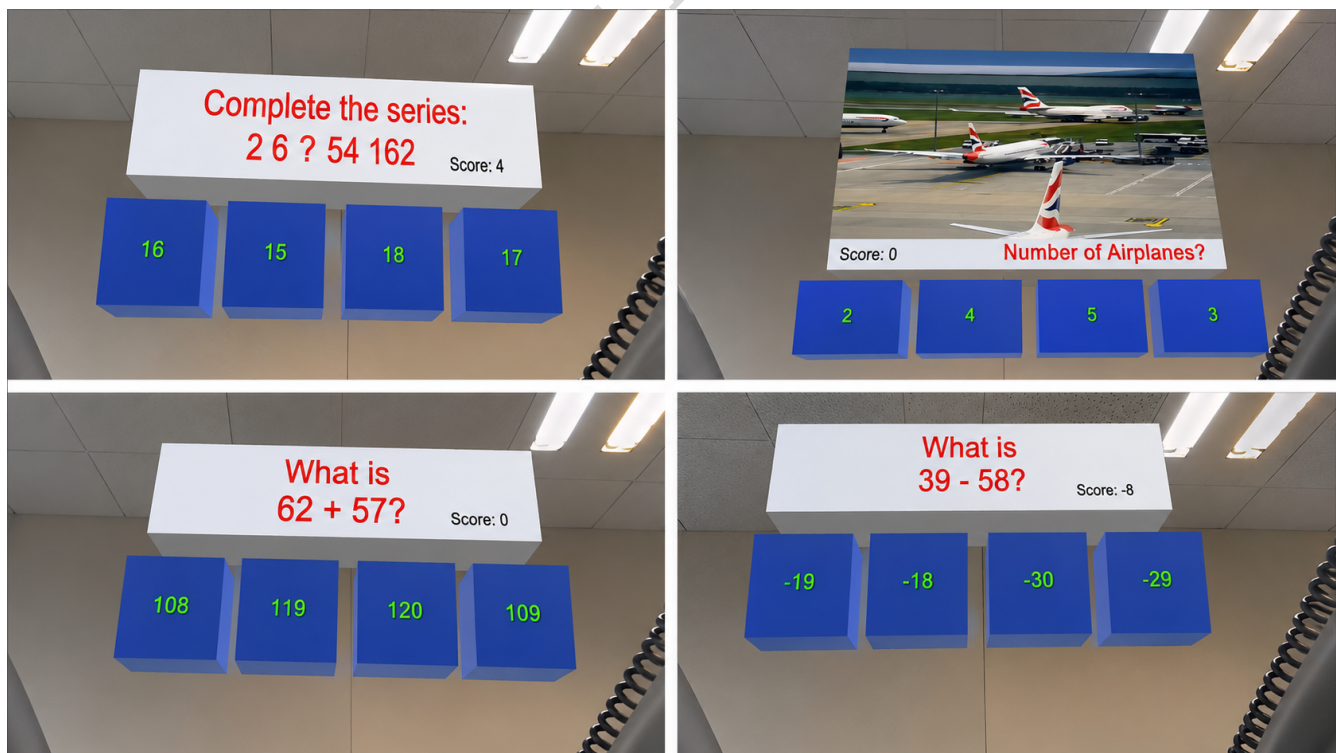


Figure 4. This figure shows a set of four exemplary games in VR environment designed to distract the participants during walking on treadmill and increase their cognitive load. The games include solving summations and subtractions, completing a math series, and counting specific objects in pictures.

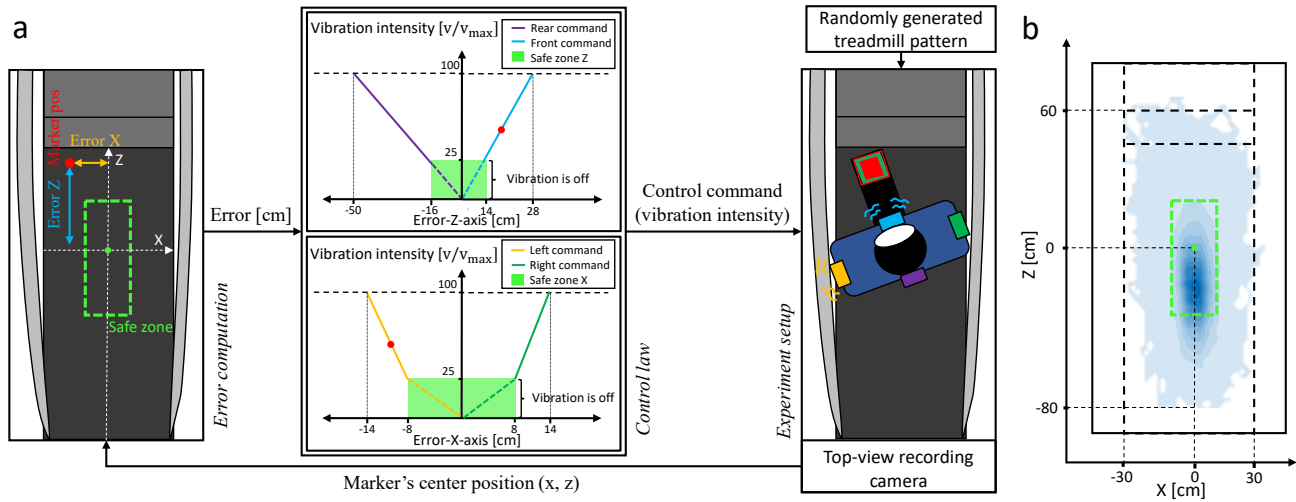
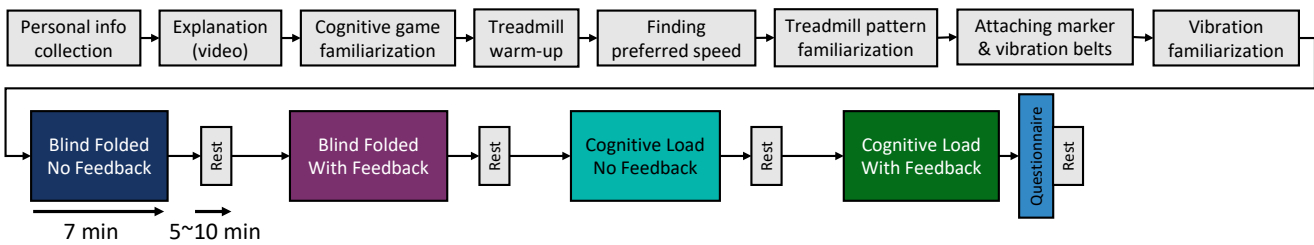


Figure 5. (a) Schematic illustration of the vibrotactile feedback control strategy. The camera and image-processing subsystem continuously tracks the 2D position of the marker's center (X, Z) relative to the center of the predefined safe zone. No vibration is delivered while the marker remains within the safe zone. Once the marker leaves this region from front, rear, right, and left the corresponding vibrotactile feedback is activated, and its intensity increases proportionally to the error (distance between the marker position and the center of safe zone), guiding the participant back toward the safe zone. (b) Contour plot of the spatial distribution of the 2D positions of the marker for all participants across all scenarios.

Day 1 - familiarization



Day 2 - main experiment

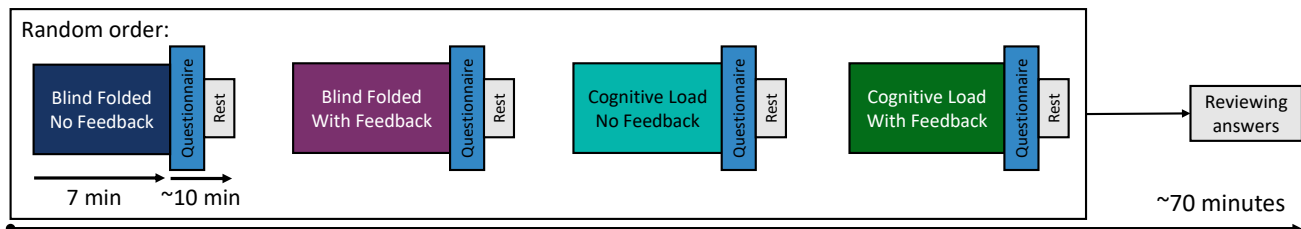


Figure 6. The flowchart of experiment protocol for familiarization and the main experiment sessions.

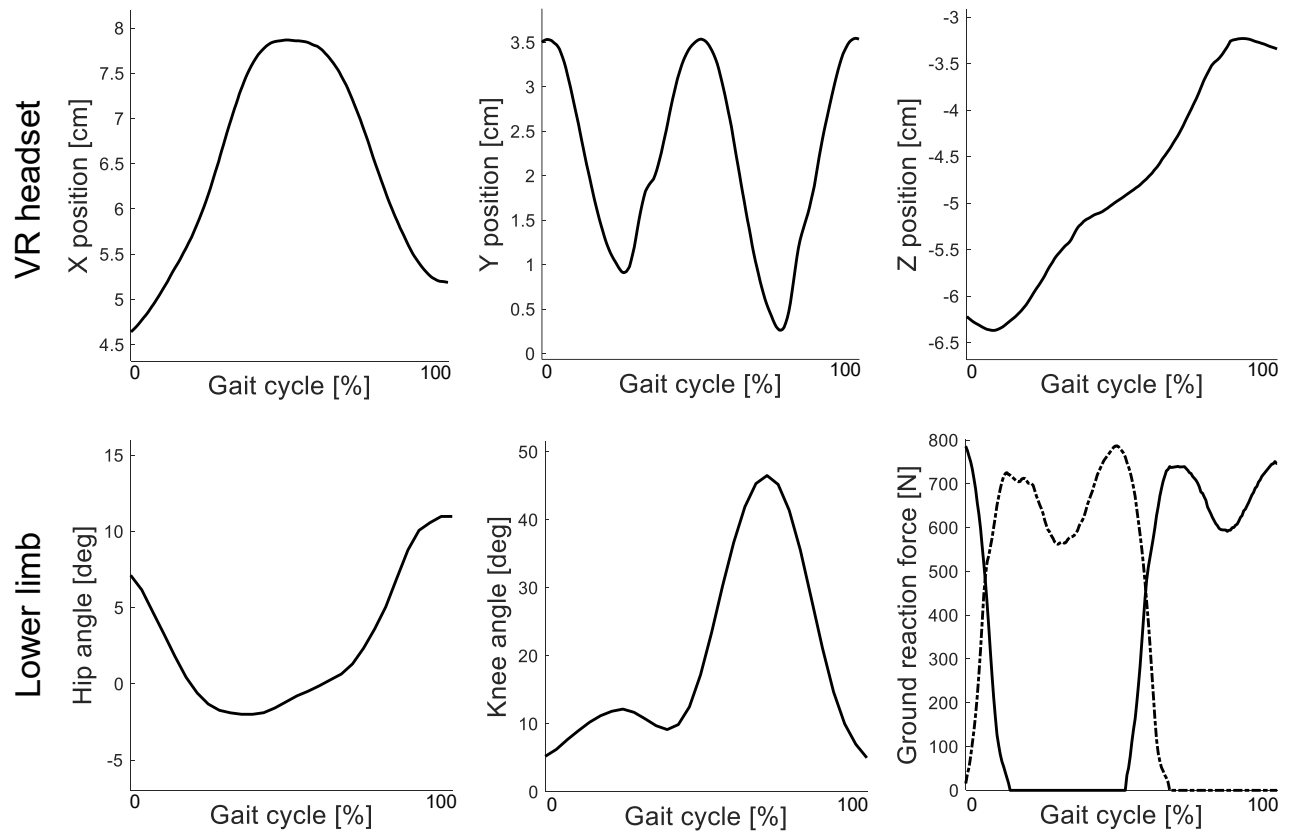


Figure 7. Some lower joint trajectories in sagittal plane extracted by MediaPipe from side-view videos, ground reaction force from force plates, and positions of the VR headset in 3 axis.

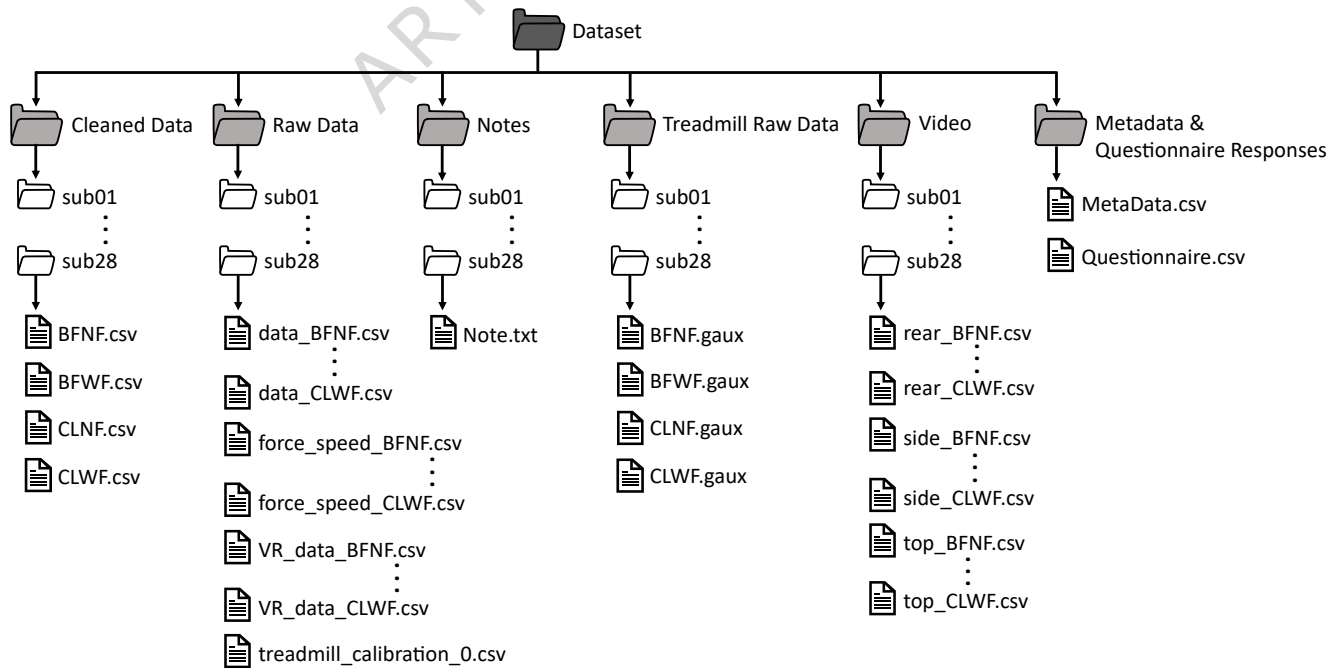


Figure 8. The figure illustrates the dataset structure.

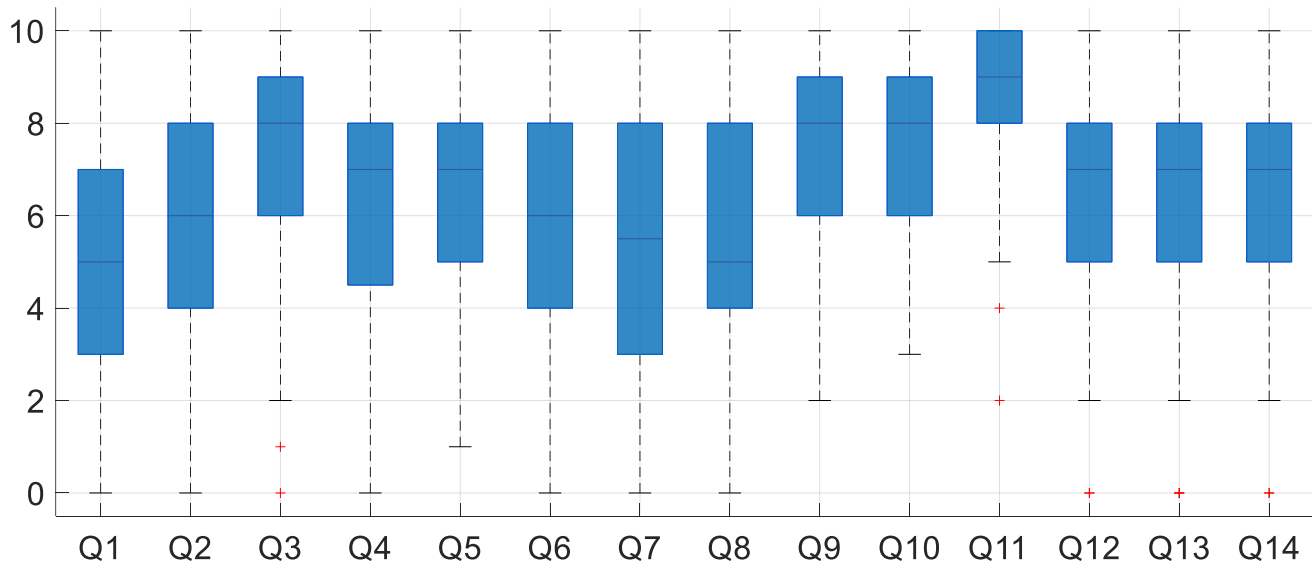


Figure 9. This figure illustrates how participants rated each question for all scenarios. Participants were asked to rate the questions by a distinct number from 0 to 10. The complete list of questions is provided in Table 2.

Table 2. List of questions used in the participant questionnaire.

Questions common in all scenarios	Questions only in the vibrotactile scenarios
Q1. How much difficulty did you experience?	Q9. To what extent was the vibrotactile feedback effective in maintaining your stability?
Q2. How comfortable did you feel?	Q10. To what extent did you pay attention to the vibrotactile feedback to maintain your stability?
Q3. How safe did you feel?	Q11. To what extent did you trust the vibrotactile feedback to maintain your stability?
Q4. How stable did you feel?	Q12. To what extent did the vibrotactile feedback help reduce the fear of instability?
Q5. To what extent did the perturbations resemble real-life conditions?	Q13. To what extent did the vibrotactile feedback help reduce your physical effort required to maintain stability?
Q6. Compared to normal walking, how much fear did you feel?	Q14. To what extent did the vibrotactile feedback help reduce your cognitive effort required to maintain stability?
Q7. Compared to normal walking, how much physical effort did you experience?	
Q8. Compared to normal walking, how much cognitive effort did you experience?	

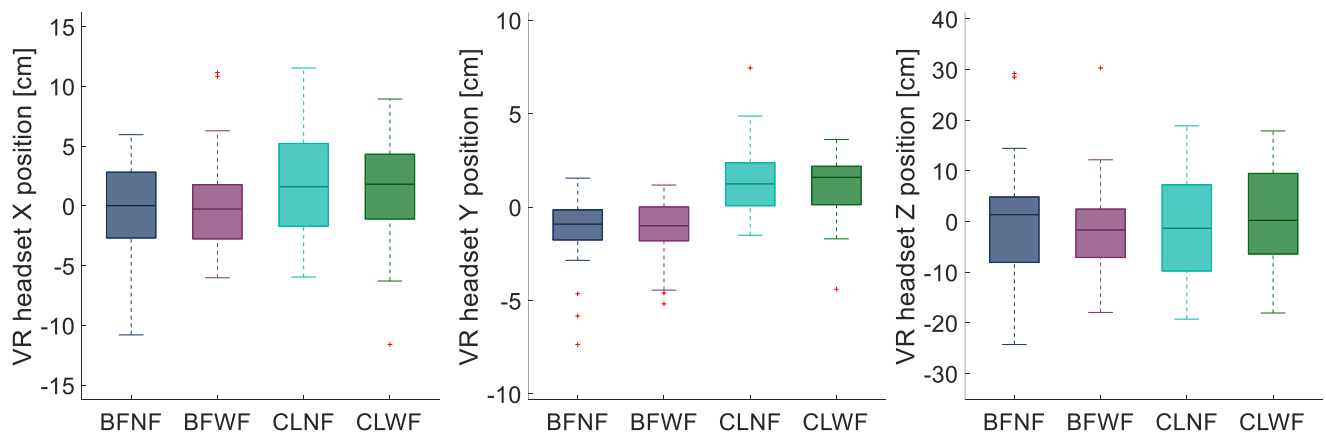


Figure 10. VR headset position (X, Y, Z) distribution of all participants across four experimental scenarios: blindfold without vibrotactile feedback (BFNF), blindfold with vibrotactile feedback (BFWF), VR-induced cognitive load without vibrotactile feedback (CLNF), VR-induced cognitive load with vibrotactile feedback (CLWF).

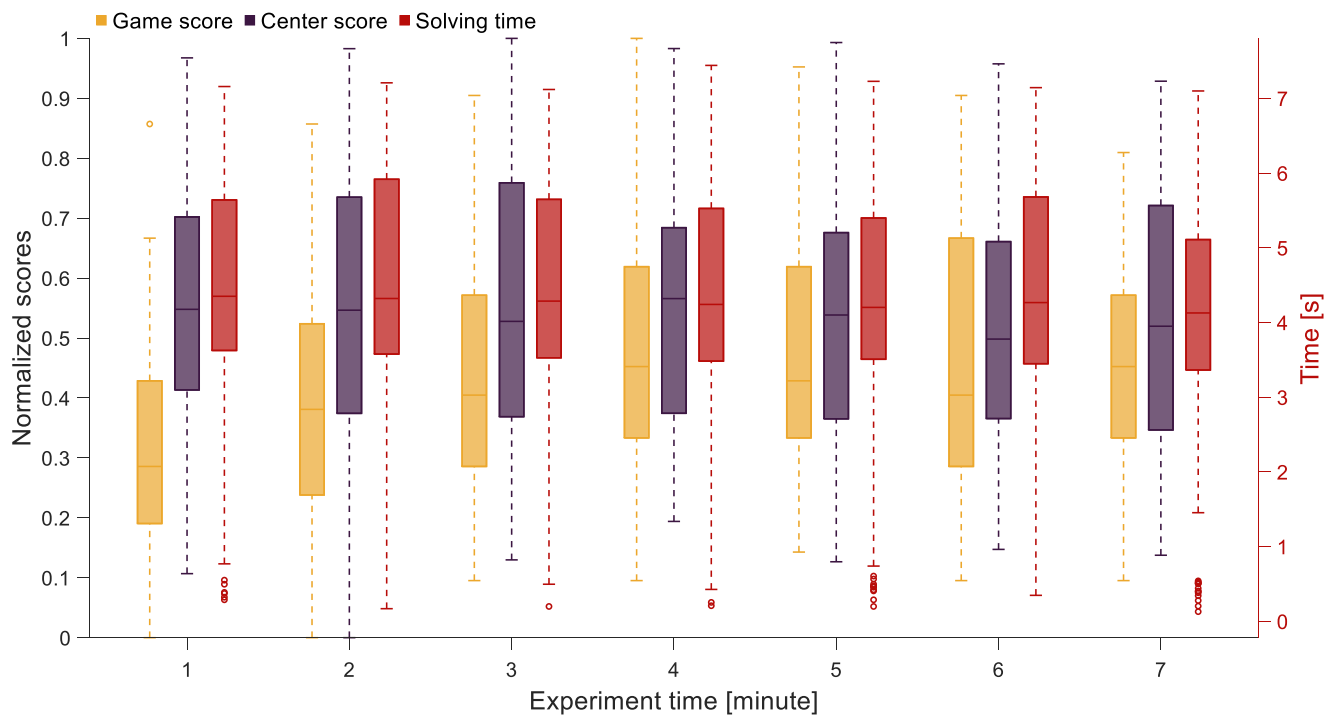


Figure 11. Distribution of the 3 subset of data across the 7-minute test. The right y-axis corresponds to the original scale for **solving time**, which represents the time spent by subjects successfully solving questions from the moment the question was instantiated until they provided a response. The left y-axis tracks the normalized scale (scaled from 0 to 1) for the two other subsets of data. Here, **game score** displays the distribution of VR game scores received by all subjects during each independent minute, and **center score** represents the participant's ability to maintain position near the central area of the treadmill during each independent minute.